Hackathon Problem Statement — Early Warning for Seniors Living at Home

Samarth Life Management Pvt Ltd

Context

India has over 150 million people aged 60+, and a growing share live alone or with only a spouse while their children work in other cities or abroad. Most health decline in this group is gradual: a missed medicine, poorer sleep, eating less, a fall that goes unreported, a change in mood. Each signal is small, and they are usually noticed only after an emergency.

Samarth Life Management supports 40,000+ seniors across 350+ cities through Care Coordinators who make regular check-in calls and visits. Coordinators each look after many families, and today they rely on memory, call notes and judgement to decide who needs attention first. We want to see how AI and data can help them spot decline days or weeks earlier.

Problem statement

Build an AI-powered early-warning system that detects health and wellbeing decline in seniors living at home, and tells a Care Coordinator who needs attention today, why, and what to do next.

The system should combine everyday signals (check-in call notes, vitals, medication adherence, activity and family inputs) into a daily risk view per senior. It must explain each alert in plain language and recommend a concrete action, such as a call, a doctor visit or a medicine review.

Users

User	What they need
Care Coordinator	A ranked daily list of seniors to prioritise, with the reason for each alert and a suggested action
Senior	Simple ways to share how they feel: a voice note, a missed-call check-in or a one-tap reply, in Hindi or English
Family member	A short weekly wellbeing summary and an immediate alert only when something is serious
Operations manager	A view of alert volumes, response times and outcomes across a city or team

Core requirements

Every team must deliver these four capabilities in a working prototype.

1. **Signal ingestion.** Accept at least three data types per senior: structured vitals (BP, sugar, SpO2, weight), medication logs and free-text or voice call notes.
2. **Unstructured understanding.** Use NLP or speech-to-text to extract health cues from call notes, for example pain, confusion, low appetite, loneliness or a fall. Notes may mix Hindi and English.
3. **Risk scoring.** Produce a daily risk score per senior using a rules baseline plus at least one ML or anomaly-detection model. Changes from the senior's own baseline matter more than population averages.
4. **Explainable action.** Show the top reasons behind each score and a recommended next step, in a Care Coordinator dashboard sorted by urgency.

Data

No real patient data will be shared. Teams generate a synthetic cohort of at least 200 seniors over 90 days, with realistic noise, missing entries and a few planted decline cases, and document how they built it.

Teams may also use public datasets to train or validate components:

- PhysioNet open datasets for vitals and time-series patterns
- Public fall-detection and human-activity datasets (accelerometer data)
- Open Hindi–English (code-mixed) text corpora for NLP
- Open speech-to-text models that support Indian languages

Cite every dataset and model used.

Constraints

- **Not a diagnosis tool.** The system flags risk and routes it to a human. It must never tell a senior or family member they have a condition.
- **Low-tech seniors.** Many seniors use basic phones, have poor eyesight or hearing, and do not use apps. Any senior-facing input must work over voice or a simple call.
- **Alert fatigue.** A Care Coordinator handles many families. False alarms erode trust, so precision matters as much as recall.
- **Privacy.** Treat all data as sensitive health data: role-based access, consent for family sharing, and no personal data sent to external services without masking.
- **Low cost.** The solution should run on modest cloud resources and remain affordable at 10,000+ seniors.

Stretch goals

Optional, for extra credit.

- A voice check-in bot that calls the senior in Hindi, asks 3–4 wellbeing questions and logs the answers
- An LLM-generated weekly family summary written in warm, plain language
- Fall or inactivity detection from phone or wearable sensor data
- A feedback loop where Coordinators mark alerts as useful or not, and the model improves from it

- Medication interaction or missed-dose pattern flags

Deliverables

1. A working prototype, demonstrated live on the synthetic cohort
2. Source code in a public or shared Git repository, with a README to run it
3. A short write-up (max 4 pages): architecture, models used, how the synthetic data was built, and results
4. Model metrics on the planted decline cases: precision, recall and how many days early decline was caught
5. A 5-minute pitch and demo

Evaluation criteria

Criterion	Weight	What judges look for
Technical depth	25%	Sound use of ML, NLP and data engineering; sensible model choices for the data size
Real-world usefulness	25%	Would a Care Coordinator act on this tomorrow? Clear priorities, low false alarms
Explainability	15%	Every alert shows why it fired, in plain language
Accessibility for seniors	15%	Works for low-tech, Hindi-speaking users with no app
Privacy and safety	10%	Sensible handling of health data; no diagnostic claims
Presentation	10%	Clear demo, honest about limitations